

Carter Johnson - Project Reflection

In this section of the module we were tasked with creating a prototype of a system that would improve quality of life by incorporating all that we had learned so far. This means synthesizing the moisture detectors that we created with the arduinos and breadboard elements to make a single system that would be responsive to the environment. My partner and I decided to create an automatic windshield wiper and headlight system that would read moisture and light values and then respond based on that information. Our inputs would be the light detector, moisture detector, and button. Our outputs would be the LEDs (headlights) and the servos (windshield wipers).

We began by setting up all pieces to the project individually on the breadboard, creating diagnostics tests to ensure that each piece works and is compatible with the arduino. We used Google Gemini to help us with code development. After we had all of our necessary inputs and outputs working correctly, we had Gemini synthesize the code into a program that would incorporate all of our components properly. We generally did not have a lot of issues with our code, as we were very specific with the information that we gave to Gemini. We also made sure to ask a lot of questions such as “is this possible?” and “is there anything else that we need to clear up?” which helped to ensure that there was not a lot of ambiguity with what we were asking for. This led to a reliable and consistent code that worked as we desired. However, we did run into one issue: we could not get our moisture detector to work consistently. This was not an issue with code, as our program was reading values. I believe that it was an issue with either the circuit design or the actual creation of the moisture sensor. This led us to have to change our threshold values regularly in order to get solid results. All other pieces to the project were a complete

success. We created a working prototype that would detect low levels of light and turn on LEDs in response. We also had a manual aspect to the windshield wipers in which you could press the button and get them working. The button also worked extremely effectively.

In this module, I learned a lot about the design process and how to troubleshoot effectively. We found the most success when we isolated individual pieces and tested them to determine whether they were working properly. I also learned a lot about how to work in a team and delegate tasks strategically. My partner and I both split tasks to tailor to each of our strong suits. She is better at circuit design than me, and I am better at code development than her, so we each took those tasks and collaborated when we were stuck on pieces. Ultimately, this module taught me how to bring an idea to life, from the ground up.